



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal to SRI VENKATESA TRADERS

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

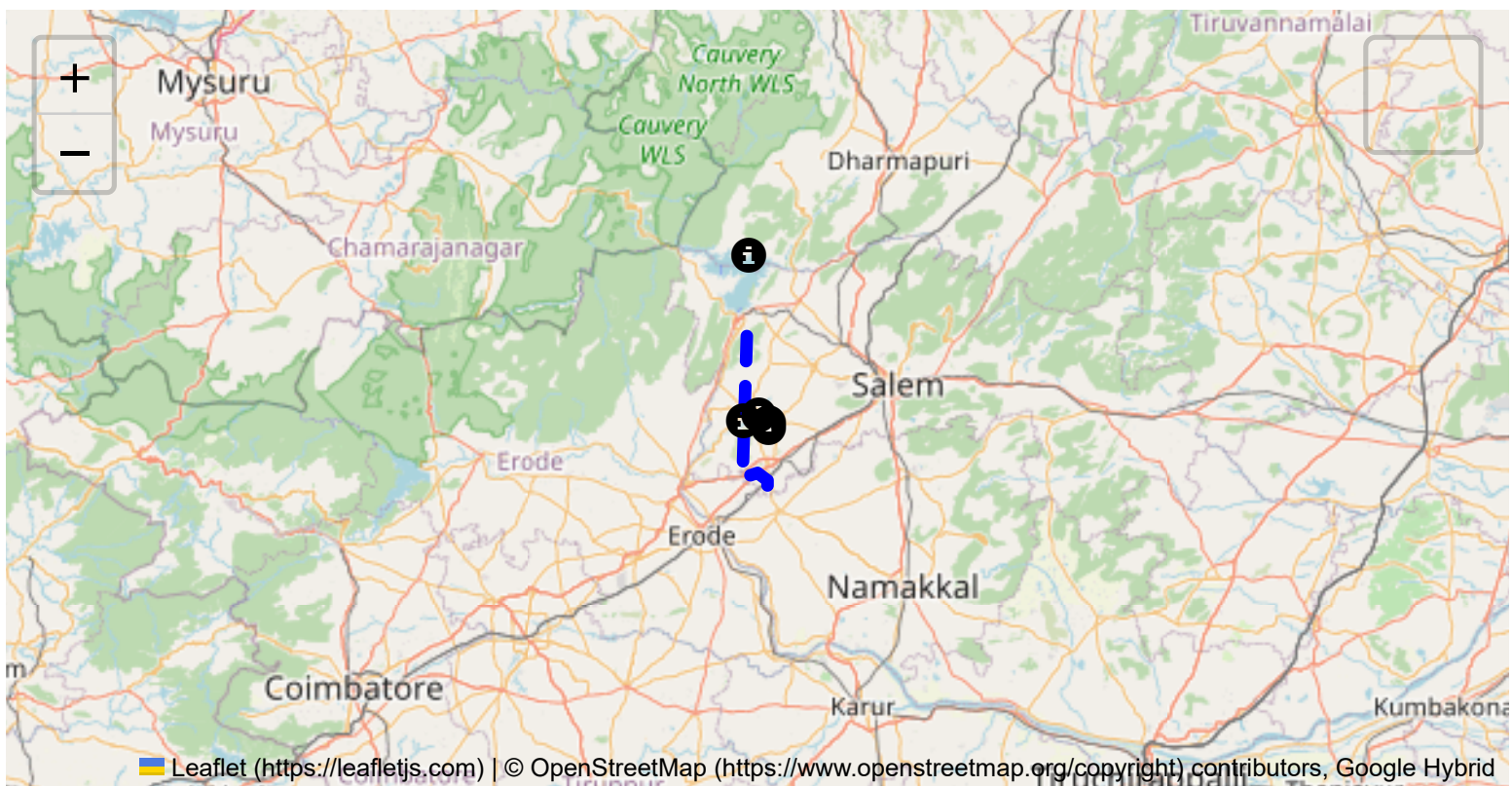
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

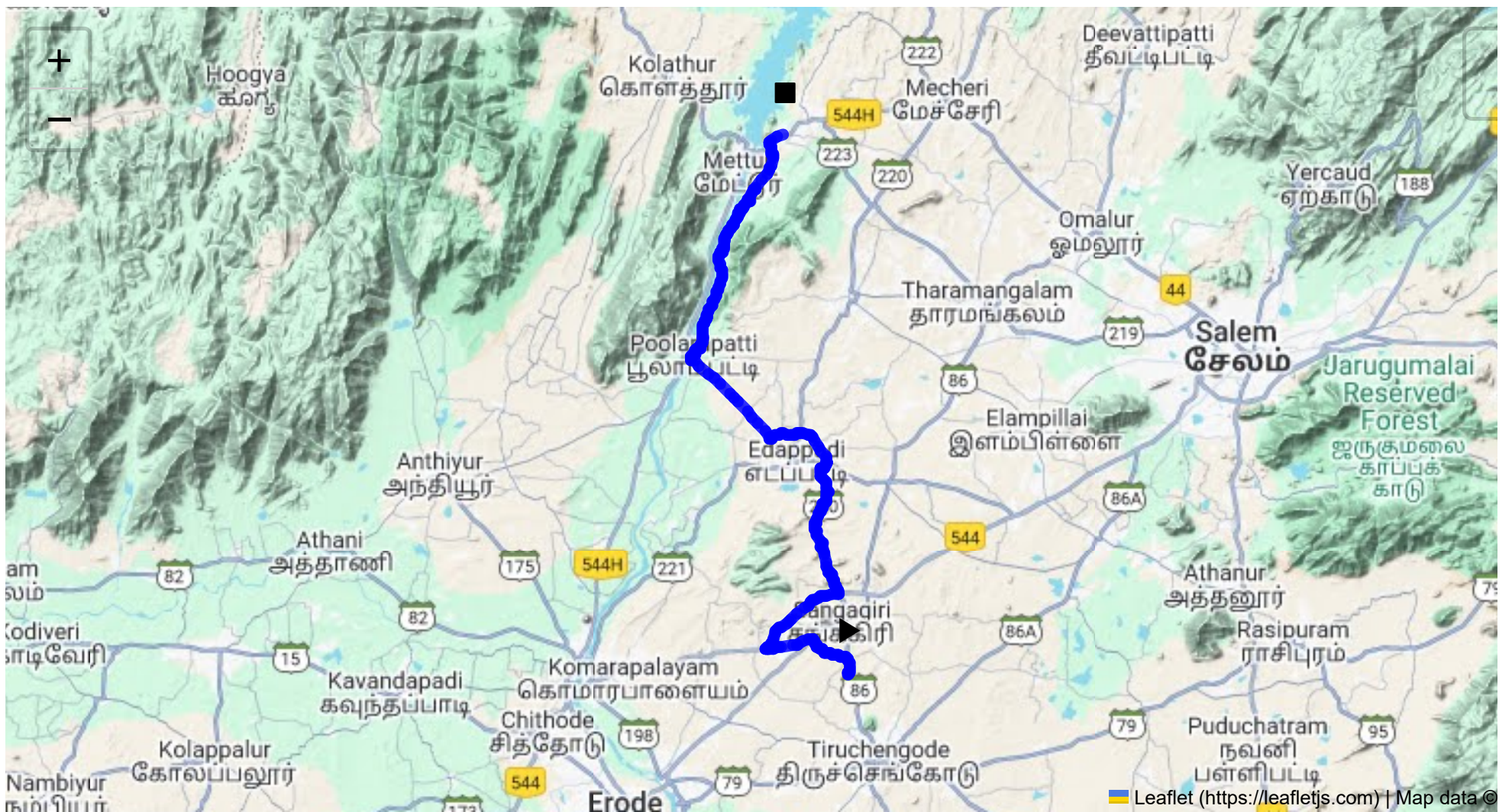
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 61.59 km
Estimated Duration: 1.5 hours
Adjusted Duration (Heavy Vehicle): 1.8 hours
Start: (11.4381, 77.8734)
End: (11.8073, 77.8282)



Welcome to the Journey Risk Management Study

Route Map Overview: The route from Sangagiri to S.V. Traders & Co. in P.N. Patti primarily follows regional roads and highways. It begins in Sangagiri with a drive south-eastward, joining Sankari RS - Erode Privu Road before merging onto Bhavani Main Road. The journey covers urban, semi-urban, and rural landscapes. The total distance is about 61.59 kilometers, and the travel time is approximately 1 hour and 46 minutes for heavy vehicles.

Weather Conditions and Hazards: Tamil Nadu generally experiences a tropical climate. During the summer months, high temperatures can affect vehicle cooling systems, while during monsoons (June to September), heavy rains might lead to water-logged roads and slippery surfaces, particularly in rural stretches.

Traffic Patterns: Typical congestion areas include urban segments near Sangagiri and closer to Salem Main Road in P.N. Patti. Peak traffic hours are usually in the morning (8-10 AM) and evening (6-8 PM). Congestion is more prevalent during local festivals or market days.

Road Quality and Infrastructure: The road quality along the route varies, with some sections of the Bhavani Main Road being prone to potholes and uneven surfaces. There are periodic highway repair projects which might cause temporary diversions or slow down traffic.

Emergency Routes: For emergencies, detours on smaller rural roads are possible, though these might not be ideal for larger vehicle transport. Alternatively, taking nearby state highways can offer faster relief from traffic jams or blockages.

Local Regulations on Hazmat Transport: Tamil Nadu has specific regulations on the transport of hazardous materials, including restrictions on timing (avoiding school zones during operational hours) and ensuring appropriate vehicle markings and documentation.

Historical Incidents: There have been occurrences of overturned trucks and minor hazardous spills, especially along heavily trafficked segments of the route. Awareness of local traffic norms and ongoing construction activities helps mitigate these risks.

Environmental Considerations: Sensitive areas include any stretches passing by agricultural zones or water bodies where spillage can have pronounced negative impacts. Awareness of local biodiversity is crucial to ensuring minimal disruption.

Communication Coverage: The majority of the route is well-covered with mobile network services. However, there might be brief dead zones, particularly in remote rural parts. It is advisable to plan communications accordingly.

Emergency Response Times: Response times vary by location; urban areas near major towns may see response times as short as 15-30 minutes, but more remote rural sections could experience delays of up to an hour for emergency services.

Overall Risk Assessment: The route poses moderate risks due to varying road conditions, potential weather impacts, and congestion. However, with appropriate preparation, adherence to regulations, and maintaining communication, these risks can be effectively managed. It is advisable for drivers to stay updated on weather forecasts, traffic news, and road conditions via reliable sources while planning their journey.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.58300, 77.85813	15 KM/Hr
1	Roundabout	High	11.60019, 77.81968	15 KM/Hr
2	Roundabout	High	11.80274, 77.81841	15 KM/Hr
3	Turn	Medium	11.43822, 77.87348	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44029, 77.87544	15 KM/Hr
6	Turn	Medium	11.44900, 77.87406	30 KM/Hr
7	Turn	Medium	11.45357, 77.85789	30 KM/Hr
8	Turn	High	11.45352, 77.85698	15 KM/Hr
9	Turn	Medium	11.45847, 77.85140	30 KM/Hr
10	Turn	Medium	11.45898, 77.85135	30 KM/Hr
11	Turn	Medium	11.46303, 77.84945	30 KM/Hr
12	Turn	Medium	11.46314, 77.84928	30 KM/Hr
13	Turn	High	11.45542, 77.81725	15 KM/Hr
14	Turn	High	11.45634, 77.81673	15 KM/Hr
15	Blind Spot	Blind Spot	11.49397, 77.86740	10 KM/Hr
16	Turn	Medium	11.49404, 77.86724	30 KM/Hr
17	Turn	Medium	11.50866, 77.86053	30 KM/Hr
18	Turn	Medium	11.52587, 77.85524	30 KM/Hr
19	Turn	Medium	11.54750, 77.85273	30 KM/Hr
20	Turn	Medium	11.55039, 77.85511	30 KM/Hr
21	Turn	Medium	11.55481, 77.85639	30 KM/Hr
22	Turn	High	11.57893, 77.85430	15 KM/Hr
23	Turn	Medium	11.57903, 77.85441	30 KM/Hr
24	Turn	Medium	11.58022, 77.85744	30 KM/Hr
25	Turn	High	11.65398, 77.76444	15 KM/Hr
26	Turn	High	11.65608, 77.76465	15 KM/Hr
27	Turn	High	11.65623, 77.76451	15 KM/Hr
28	Turn	High	11.65760, 77.76513	15 KM/Hr
29	Turn	High	11.65764, 77.76613	15 KM/Hr
30	Turn	High	11.66103, 77.76757	15 KM/Hr
31	Turn	High	11.66090, 77.76849	15 KM/Hr
32	Turn	Medium	11.66123, 77.76890	30 KM/Hr
33	Turn	High	11.66122, 77.76992	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
34	Turn	High	11.66386, 77.77083	15 KM/Hr
35	Turn	High	11.66386, 77.77125	15 KM/Hr
36	Turn	Medium	11.66400, 77.77136	30 KM/Hr
37	Turn	High	11.66857, 77.77231	15 KM/Hr
38	Turn	High	11.66873, 77.77191	15 KM/Hr
39	Turn	High	11.66911, 77.77183	15 KM/Hr
40	Turn	Medium	11.66947, 77.77209	30 KM/Hr
41	Turn	Medium	11.72539, 77.78362	30 KM/Hr
42	Turn	Medium	11.72628, 77.78415	30 KM/Hr
43	Turn	High	11.72745, 77.78411	15 KM/Hr
44	Turn	Medium	11.72759, 77.78433	30 KM/Hr
45	Turn	Medium	11.72751, 77.78550	30 KM/Hr
46	Turn	Medium	11.72785, 77.78582	30 KM/Hr
47	Turn	Medium	11.73941, 77.78811	30 KM/Hr
48	Turn	Medium	11.73971, 77.78838	30 KM/Hr
49	Turn	Medium	11.74000, 77.78938	30 KM/Hr
50	Turn	Medium	11.75274, 77.79710	30 KM/Hr
51	Turn	Medium	11.75599, 77.79872	30 KM/Hr
52	Turn	High	11.75676, 77.79842	15 KM/Hr
53	Turn	High	11.75724, 77.79860	15 KM/Hr
54	Turn	Medium	11.75752, 77.79996	30 KM/Hr
55	Turn	Medium	11.76229, 77.80500	30 KM/Hr
56	Turn	High	11.76312, 77.80533	15 KM/Hr
57	Turn	Medium	11.76347, 77.80512	30 KM/Hr
58	Turn	High	11.76402, 77.80425	15 KM/Hr
59	Turn	Medium	11.76436, 77.80420	30 KM/Hr
60	Turn	Medium	11.76990, 77.80761	30 KM/Hr
61	Turn	High	11.77028, 77.80821	15 KM/Hr
62	Turn	High	11.77045, 77.80821	15 KM/Hr
63	Turn	Medium	11.79440, 77.82114	30 KM/Hr
64	Turn	Medium	11.79572, 77.81991	30 KM/Hr
65	Turn	Medium	11.80213, 77.81890	30 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
1	hospital	Chinnakavur Hospital	11.7830314, 77.8206836	30 km/h	Medium
2	clinic	Suresh Physiotherapy Clinic	11.804642, 77.821786	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
0	school	KRP Matric. Hr. Sec School	11.4546193, 77.8142445	30 km/h	Medium
3	school	Vaitheeshwara Hr. Sec. School	11.8068419, 77.8242984	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.58300, 77.85813



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.60019, 77.81968



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.80274, 77.81841



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.44029, 77.87544



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.44900, 77.87406



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45357, 77.85789



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45352, 77.85698



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.45847, 77.85140



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.45898, 77.85135



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46303, 77.84945



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.46314, 77.84928



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45542, 77.81725



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.45634, 77.81673



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.49397, 77.86740



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.49404, 77.86724



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.50866, 77.86053



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.52587, 77.85524



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.54750, 77.85273



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55039, 77.85511



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.55481, 77.85639



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.57893, 77.85430



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.57903, 77.85441



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.58022, 77.85744



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65398, 77.76444



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65608, 77.76465



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.65623, 77.76451



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65760, 77.76513



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.65764, 77.76613



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66103, 77.76757



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66090, 77.76849



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66123, 77.76890



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66122, 77.76992



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66386, 77.77083



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66386, 77.77125



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66400, 77.77136



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.66857, 77.77231



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66873, 77.77191



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.66911, 77.77183



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.66947, 77.77209



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72539, 77.78362



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72628, 77.78415



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.72745, 77.78411



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.72759, 77.78433



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72751, 77.78550



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.72785, 77.78582



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.73941, 77.78811



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.73971, 77.78838



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.74000, 77.78938



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75274, 77.79710



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.75599, 77.79872



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.75676, 77.79842



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.75724, 77.79860



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.75752, 77.79996



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76229, 77.80500



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.76312, 77.80533



Risk Type: Turn
Risk Level: Medium
Speed Limit: 30 KM/Hr
Coordinates: 11.76347, 77.80512

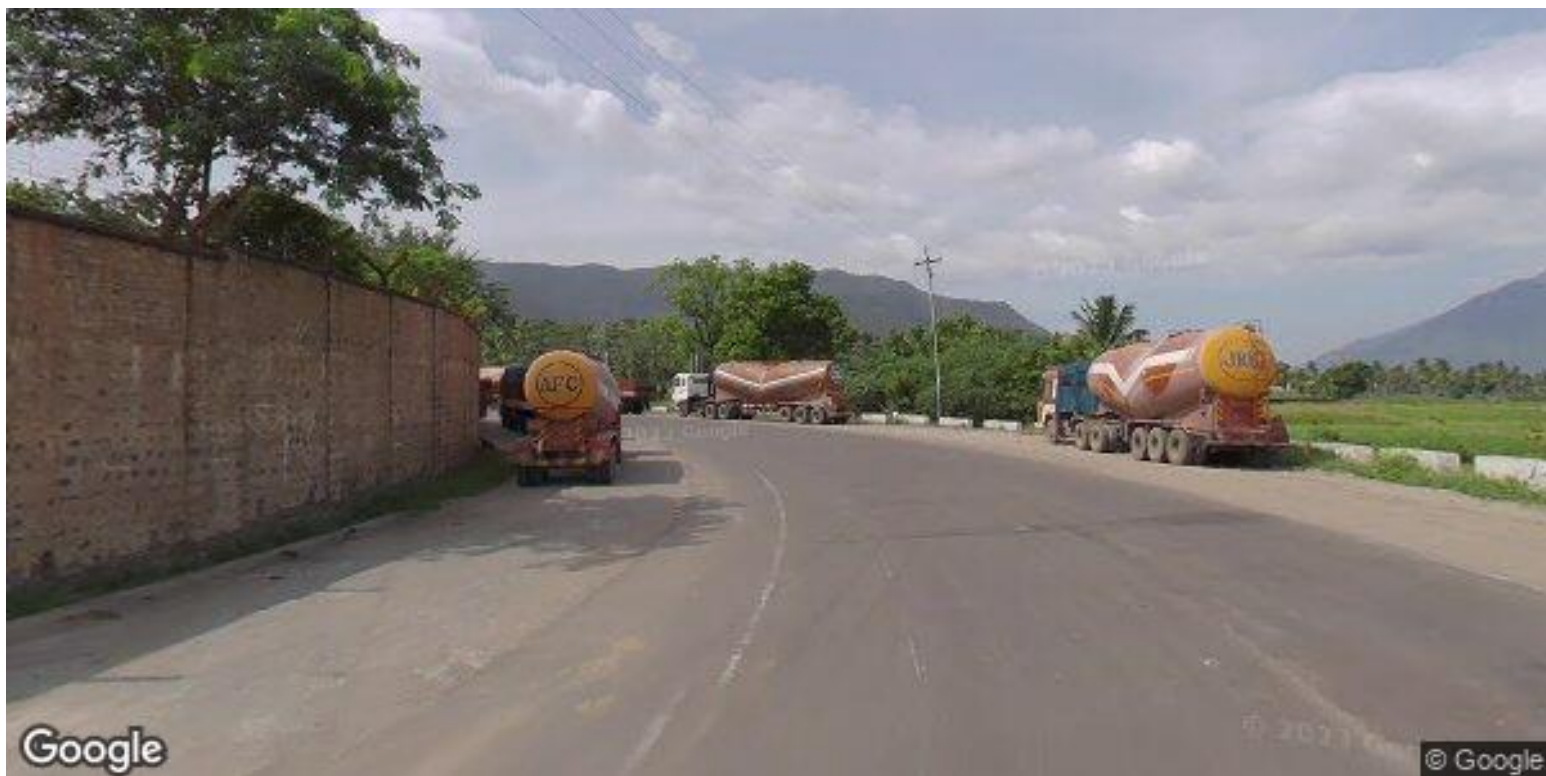


Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.76402, 77.80425



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76436, 77.80420



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.76990, 77.80761



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.77028, 77.80821



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.77045, 77.80821



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.79440, 77.82114



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.79572, 77.81991



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.80213, 77.81890
